

1. Administrative & Study Identification

Full Study Title: A Phase IV Open-Label Study Evaluating the Effectiveness and Safety of Risdiplam Administered as an Early Intervention in Pediatric Patients With Spinal Muscular Atrophy After Gene Therapy **Short Title/Acronym:** HINALEA 1 **Protocol Number:** BN44620 (EU CT: 2023-504508-26-00) **NCT Number:** NCT05861986 **Sponsor Name:** Hoffmann-La Roche (Roche) **Investigational Product:** Risdiplam (Evrysdi) | **ATC Code:** M09AX10 **Phase of Development:** Phase IV **Trial Status:** Recruiting **Medical Monitor:** Hoffmann-La Roche Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the effectiveness and safety of risdiplam administered as an early intervention in pediatric participants with spinal muscular atrophy (SMA) and 2 SMN2 copies who have previously received onasemnogene abeparvovec (gene therapy). **Secondary Objectives:** To evaluate the achievement of motor milestones (measured by BSID-III and PDMS-3), swallowing and feeding ability (measured by p-FOIS), and the long-term safety and tolerability of risdiplam.

2.2 Background & Rationale Currently, there are reported cases of risdiplam being started after SMA gene therapy (onasemnogene abeparvovec), yet there is no standard way this is done. Some doctors add risdiplam early after gene therapy, and some start risdiplam later when the patient may not be responding well. This trial is studying the effects, good or bad, of risdiplam given as an early intervention after gene therapy to evaluate if it may provide additional benefit and is safe for children genetically diagnosed with SMA.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Single-arm **Blinding:** Open-label **Randomization:** N/A

3.2 Planned Enrollment & Duration Target \$N\$: 28 estimated patients **Number of Sites:** 16 locations (Multicenter, including US sites) **Estimated Study Start:** 30-May-2024 **Estimated Primary Completion:** 31-Mar-2028

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. <2 years of age at the time of informed consent. 2. Confirmed diagnosis of 5q-autosomal recessive SMA, including genetic confirmation of homozygous deletion or compound heterozygosity predictive of loss of function of the Survival of Motor Neuron 1 (SMN1) gene. 3. Confirmed presence of two SMN2 gene copies as documented through laboratory testing. 4. Administration of onasemnogene abeparvovec pre-symptomatically or post-symptomatically. 5. Has received onasemnogene abeparvovec for SMA no less than 13 weeks, but not more than 30 weeks, prior to enrollment. 6. If treated with risdiplam prior to onasemnogene abeparvovec, risdiplam treatment must not have exceeded 3 weeks and must be discontinued 1 day prior to onasemnogene abeparvovec administration. 7. Has, in the opinion of the investigator, not experienced a clinically significant decline in function from the time of onasemnogene abeparvovec administration.

4.2 Exclusion Criteria 1. Previous or current enrolment in an investigational study prior to initiation of study treatment. 2. Any unresolved standard-of-care laboratory abnormalities per the onasemnogene abeparvovec prescribing information. 3. Concomitant or previous administration of an SMN2-targeting antisense oligonucleotide. 4. Concomitant or previous use of an anti-myostatin agent. 5. Participants requiring invasive ventilation or tracheostomy. 6. Participants requiring awake non-invasive ventilation or with awake hypoxemia (Arterial Oxygen Saturation [SaO₂] <95%) with or without ventilator support. 7. Presence of feeding tube and an OrSAT score of 0. 8. Hospitalization for pulmonary event within the last 2 months, or any planned hospitalization at the time of screening. 9. Any major illness requiring hospitalization within 1 month before the screening examination or any febrile illness within 1 week prior to screening and up to first dose administration.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Risdiplam administered orally once daily (dose adapted for weight and age). **Route of Administration:** Oral (liquid given by mouth or by feeding tube). **Duration of Treatment:** 72 weeks (Treatment Period) followed by a 1-year Treatment Extension Period (total 120 weeks / ~2.5 years of exposure).

5.2 Concomitant & Prohibited Therapy Permitted: Standard of care treatments for SMA (excluding prohibited medications).

Prohibited: SMN2-targeting antisense oligonucleotides, anti-myostatin agents, and unresolved standard-of-care laboratory abnormalities per the onasemnogene abeparvovec prescribing information.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, Genetic confirmation of SMA, Baseline Motor Assessments
Baseline	Day 0	First dose of Risdiplam, Safety Labs, Baseline Swallowing/Feeding Evaluation
Interim	Every 12 Weeks (Q12W)	Clinic visits, Motor function evaluation (BSID-III, PDMS-3), Safety & side effect tracking
End of Study	Week 120	Final Motor and Feeding Assessments, Exit Interview, IP Accountability

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Change from Baseline in the Raw Score of Bayley Scales of Infant and Toddler Development - Third Edition (BSID-III) Gross Motor Score at 72 Weeks of Risdiplam Treatment. **Measurement Method:** Clinical assessment via standardized scales. The BSID-III gross motor scale consists of 72 items scored at 0 (unable to perform) or 1 (criteria for item achieved). A higher raw score indicates improvement.

7.2 Secondary & Exploratory Endpoints

1. Percentage of Participants With Adverse Events.
2. Percentage of Participants With Serious Adverse Events.
3. Percentage of Participants With Treatment Discontinuation Due to Adverse Events.
4. Gross and fine motor skill improvements evaluated using the Peabody Developmental Motor Scales, Third Edition (PDMS-3).
5. Swallowing and feeding ability assessed by the Pediatric Functional Oral Intake Scale (p-FOIS).

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Collection via routine clinical assessments every 12 weeks, caregiver diaries, and laboratory safety parameters. **Serious Adverse Events (SAEs):** Reported within 24 hours of site/investigator awareness. **Data Monitoring Committee (DMC):** Independent/Internal safety review mechanism organized by Hoffmann-La Roche.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Safety Set (all participants receiving ≥ 1 dose of risdiplam); Efficacy Set (participants with baseline and post-baseline effectiveness assessments). **Sample Size Justification:** Target $N=28$. This sample size is designed to provide descriptive statistics for evaluating safety and clinical effectiveness trends in a rare disease population. **Handling of Missing Data:** Data will be handled per protocol utilizing

standard techniques like Last Observation Carried Forward (LOCF) or Mixed-Effect Model Repeated Measure (MMRM) for motor scores.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Regular onsite and remote monitoring visits by sponsor representatives to assure data integrity. **Audit Trail:** eCRF locking, source data verification, and data clarification forms will be securely maintained.

11. Study Identifiers & Governance

Primary ID: BN44620 **Posting ID:** 100205503239627507 **Registry Number:** NCT05861986 **Sponsor/Lead Organization:** Hoffmann-La Roche (Roche) **Study Title:** HINALEA 1 **Official Regulatory Title:** A Phase IV Open-Label Study Evaluating the Effectiveness and Safety of Risdiplam Administered as an Early Intervention in Pediatric Patients With Spinal Muscular Atrophy After Gene Therapy **EMA Product Information:** [Evrysdi EPAR](#)

12. Clinical Framework (SMA Specific)

Primary Condition: Spinal Muscular Atrophy (SMA) **Diagnosis Threshold:** Genetically confirmed 5q-autosomal recessive SMA, 2 SMN2 copies **Medical Methodology:** Early intervention with SMN2-directed RNA splicing modifier (Risdiplam) following gene therapy (onasemnogene abeparvovec) **Optimization Target:** Motor function maintenance/improvement and feeding ability safety

13. Study Procedures & Methodology

Management Pathway: Daily oral or feeding tube administration of risdiplam at home **Education Protocol (HCP):** Site training on administration and standardized motor functioning assessments (BSID-III, PDMS-3, p-FOIS) **Education Protocol (Patient):** Parent/caregiver training on proper preparation, dosage calculation, and administration via mouth or feeding tube **Quality Audit Mechanism:** Regular clinical site monitoring and review of caregiver logs for patient adherence

14. Observation & Follow-Up Schedule

Total Duration: 120 weeks (approx. 2.5 years) **Onsite Visit Cadence:** Every 12 weeks (Week 12, 24, 36, 48, 60, 72, etc.) **Remote Monitoring Frequency:** Telehealth/phone check-ins as

specified between 12-week site visits **Checklist Review Interval:**
Caregiver dosing diaries reviewed every 12 weeks

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---	
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]					
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]					Lead
Statistician	[Name]	_____	[DD-MMM-YYYY]					